

EAST TEXAS, TX, USA

WMO: 722470

Lat: 32.385N

Lon: 94.712W

Elev: 365

StdP: 14.50

Time Zone: -6.00 (NAC)

Period: 98-19

WBAN: 03901

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.4	28.2	12.1	10.3	34.7	16.3	12.7	37.6	24.2	56.5	21.2	60.4	4.7	350	0.431

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.6	99.9	75.1	97.5	75.6	95.0	75.7	79.4	90.2	78.5	89.2	78.0	88.5	7.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.0	142.6	82.6	76.1	138.0	82.0	75.3	134.2	81.7	43.0	90.5	42.2	89.5	41.5	88.7	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.2	18.1	16.2	DB	19.2	102.4	3.8	3.1	16.5	104.6	14.3	106.4	12.2	108.2	9.5	110.4	
			WB	17.2	80.4	3.5	1.2	14.7	81.3	12.6	82.0	10.6	82.6	8.0	83.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	66.6	47.6	51.4	58.9	66.1	73.8	80.8	83.8	84.0	78.1	67.3	57.0	49.1
	DBStd	15.09	10.02	10.28	8.99	7.40	6.05	4.11	3.70	4.20	5.75	8.03	9.24	9.50
	HDD50	472	169	99	27	2	0	0	0	0	0	1	37	137
	HDD65	2098	547	393	230	75	10	0	0	0	2	71	269	501
	CDD50	6513	93	138	301	484	737	924	1046	1055	842	537	248	108
	CDD65	2665	6	12	39	107	282	474	581	590	394	142	30	8
	CDH74	28204	17	60	252	849	2478	5044	6906	7162	3993	1247	167	29
	CDH80	13273	0	4	33	173	890	2397	3579	3885	1887	403	21	1
Wind	WSAvg	7.1	7.7	8.2	8.4	8.7	7.8	6.9	6.2	5.8	5.4	6.2	7.0	7.3
Precipitation	PrecAvg	47.2	4.3	3.9	4.4	3.5	4.0	4.4	3.3	3.0	3.7	4.1	3.9	4.5
	PrecMax	70.4	8.6	9.3	9.4	13.0	9.0	14.6	11.0	7.4	9.1	15.0	11.1	10.0
	PrecMin	28.5	0.7	0.4	0.5	0.5	0.8	0.1	0.2	0.0	1.0	0.1	0.4	0.7
	PrecStd	11.2	2.0	2.1	2.0	2.6	2.3	3.3	2.3	1.8	2.3	3.4	2.5	2.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.4	79.6	83.7	88.3	93.5	99.8	102.0	104.3	100.3	90.9	83.0	77.5
		MCWB	62.5	64.1	66.8	71.2	73.5	75.4	75.6	74.3	73.1	72.4	69.5	66.4
	2%	DB	72.6	75.5	80.3	84.1	90.2	96.6	99.4	100.3	95.7	87.8	78.9	73.1
		MCWB	61.7	63.5	66.2	69.4	73.7	75.4	75.3	75.1	74.3	71.4	67.2	65.1
	5%	DB	69.1	71.9	76.7	81.7	87.7	93.4	96.6	98.0	92.8	84.6	75.0	69.9
		MCWB	61.1	62.9	63.8	68.2	73.2	75.3	76.0	75.9	74.0	70.0	65.1	63.6
	10%	DB	65.3	68.5	73.2	79.3	85.3	91.0	93.7	95.3	90.1	81.4	72.2	66.0
		MCWB	58.2	60.8	62.5	66.6	72.2	75.7	76.5	75.9	73.6	68.4	64.5	60.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.4	69.4	70.2	74.8	77.8	80.1	80.4	80.3	79.3	77.2	72.9	69.7
		MCDB	73.1	74.0	77.8	81.4	87.2	89.6	90.7	91.0	88.8	84.9	78.4	73.4
	2%	WB	65.3	67.1	68.6	72.5	76.3	78.7	79.3	79.3	77.9	75.4	70.0	67.3
		MCDB	69.6	72.3	75.6	80.2	85.3	88.5	90.4	90.8	87.2	82.6	75.2	71.9
	5%	WB	63.3	64.9	67.1	71.0	75.1	77.9	78.5	78.4	76.9	73.6	68.1	64.7
		MCDB	67.5	69.9	73.5	78.7	83.8	87.9	89.3	89.7	86.0	80.1	73.3	68.9
	10%	WB	59.3	62.3	65.0	69.4	73.9	77.0	77.8	77.7	75.9	71.8	65.9	61.3
		MCDB	63.7	67.3	71.5	76.4	82.3	86.8	88.5	88.8	84.7	77.6	71.0	65.3
Mean Daily Temperature Range	5% DB	MDBR	20.5	20.0	20.9	21.3	19.5	18.7	19.2	20.6	21.4	22.2	21.3	20.1
		MCDBR	24.5	23.0	23.0	22.4	21.0	21.8	22.7	24.1	24.2	24.1	23.1	23.3
		MCWBR	15.3	13.9	11.8	10.5	8.0	6.0	5.2	5.7	7.2	10.1	12.7	15.7
	5% WB	MCDBR	19.4	19.0	19.4	18.2	18.2	17.9	18.6	19.8	18.9	19.0	18.5	20.0
		MCWBR	15.3	14.2	12.3	10.2	8.5	6.5	6.0	5.8	6.5	9.8	12.5	15.6
Clear-Sky Solar Irradiance	taub	0.325	0.340	0.360	0.398	0.429	0.452	0.467	0.462	0.418	0.363	0.349	0.330	
	taud	2.504	2.468	2.424	2.321	2.287	2.264	2.258	2.263	2.348	2.479	2.466	2.501	
	Ebn at Noon	285	291	292	283	274	266	261	261	269	279	272	276	
	Edh at Noon	28	31	35	41	43	43	43	43	38	31	28	26	
All-Sky Solar Radiation	RadAvg	869	974	1366	1699	1850	1994	2017	1891	1607	1293	940	757	
	RadStd	79	161	126	106	140	165	149	152	144	176	112	98	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+322	+238
	Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page